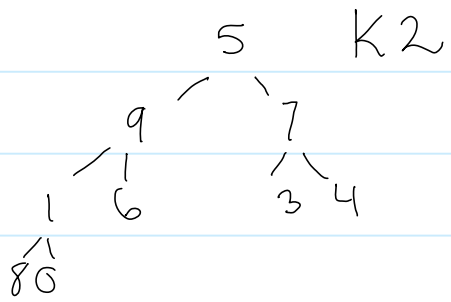
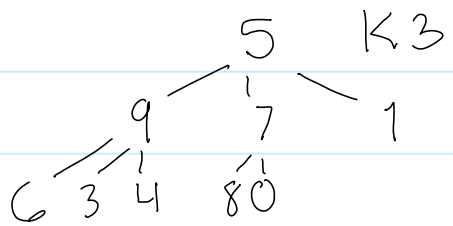
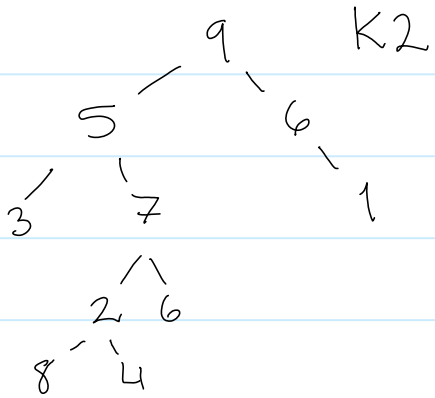
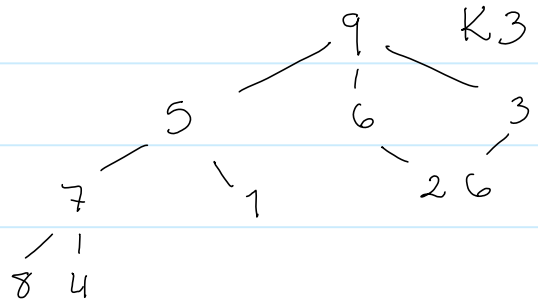
# Exercises- Week 5

Wednesday, September 18, 2024 5:05 PM




[2, 9, 6, 5, -, 1, 8, -, 3, -, 0, -, 7]

① Parent(i) is $(K + i - 2)/K$ (int division)

② $J^{Th}$ child of (i) is $K \cdot i - (K-2) + j - 1 = Ki - K + j + 1$

Binary k tree:

③ parent(i) = $i/2$ (int division)

④ Left child(i) = $2i$

⑤ Right child(i) = $2i + 1$

Tree 1:

[a, b, c, d, e, f, g, h, i, j, k, L, m] K = 3

[a, b, c, d, e, f, g, h, i, j, k, L, m] K=3

① any but 1, (1+i)/3 increases every 3 ... correct.

② for any (i | i ≥ 1), (j | 1 ≤ j ≤ 3), 3i - 3 + j + 1 = 3(i-1) + (j+1) ... correct

Tree 2:

[a, b, c, d, e, f, g, h, i, j, k, l]

③ correct ④ correct ⑤ correct.

# 1

In order: 2, 8, 6, 5, 9, 0, 7, 1, 2

Pre order: 5, 6, 2, 8, 7, 9, 0, 1, 2

Post order: 8, 2, 6, 0, 9, 2, 1, 7, 5

# 2

In order: 1, 6, 9, 3, 7, 5

Pre order: 3, 6, 1, 9, 7, 5

Post order: 1, 9, 6, 5, 7, 3

# 3

In order: 0, 1, 3, 4, 5, 6, 7, 8

Pre order: 5, 3, 1, 0, 4, 7, 6, 8

Post order: 0, 1, 4, 3, 5, 6, 8, 7

Try it again: 1, 2, 5, 6, 7, 9

How to find value inside tree: It depends on if its sorted.

What algorithm: depends where its going...?

How would removing work: make sure any kids dont get orphaned